```
jest.mock("../models/pickupRequestModel.js");
jest.mock("../models/EwasteItem.js");
jest.mock("../models/User.model.js");

const { createRequest } = require("../controllers/pickupRequestController.js");

const PickupRequest = require("../models/pickupRequestModel.js").default;
const EwasteItem = require("../models/EwasteItem.js").default;
const User = require("../models/User.model.js").default;

describe("createRequest", () => {
  let req, res;

  beforeEach(() => {
    req = {
      body: {
        ewasteItemId: "item123",
        quantity: 1,
```

```
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npm test
Ran all test suites.
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npm test

> backend@1.0.0 test
> jest

PASS tests/pickupRequestController.test.cjs
  createRequest
    √ should create a pickup request successfully (12 ms)
    √ should return 404 if ewaste item not found (2 ms)

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        1.778 s, estimated 77 s
Ran all test suites.
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend>
```

```
describe("settingsController", () => {
  describe("updateSettings", () => {
    it("should return 500 if recalculation fails", async () => {

      await updateSettings(req, res);

      expect(res.status).toHaveBeenCalledWith(500);
      expect(res.json).toHaveBeenCalledWith({
        message: "Log fetch failed",
      });
    });
  });
});
```

```
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npm test
Tests:       39 passed, 39 total
Snapshots:   0 total
Time:        3.517 s, estimated 6 s
Ran all test suites.
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npx jest tests/holidayController.test.cjs
>> npx jest tests/settingsController.test.cjs
PASS tests/holidayController.test.cjs
  holidayController
    getSriLankaHolidays
      √ should return holidays successfully for a given year and month (5 ms)
      √ should return current year if year is not provided (2 ms)
      √ should return 500 if API key is missing (1 ms)
      √ should return 500 when axios throws an API error response (1 ms)
      √ should return 500 when axios throws a normal error (1 ms)
    getNextSriLankaHoliday
      √ should return the next upcoming holiday from the current year (2 ms)
      √ should fetch next year if no upcoming holidays remain this year (1 ms)
      √ should return null if no holidays are available (2 ms)
      √ should return 500 if API key is missing (2 ms)
```

```
Backend > tests > impactLogController.test.cjs > ...
14   describe("impactLogController", () => {
361    describe("deleteImpactLog", () => {
378      it("should return 404 if impact log is not found", async () => {
387        message: "Impact log not found",
388      });
389    });
390
391    it("should return 500 on delete error", async () => {
392      req.params = { id: "log123" };
393
394      ImpactLog.findByIdAndDelete.mockRejectedValue(new Error("Delete failed"));
395
396      await deleteImpactLog(req, res);
397
398      expect(res.status).toHaveBeenCalledWith(500);
399      expect(res.json).toHaveBeenCalledWith({
400        message: "Delete failed",
401      });
```

```
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npx jest tests/impactLog.test.cjs
Run with `--passWithNoTests` to exit with code 0
In C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend
  44 files checked.
  testMatch: **/__tests__/**/*.?([mc])[jt]s?(x), **/?(*.)+(spec|test).?([mc])[jt]s?(x) - 4 matches
  testPathIgnorePatterns: \\node_modules\\ - 44 matches
  testRegex:  - 0 matches
Pattern: tests/impactLog.test.cjs - 0 matches
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npx jest tests/impactLogController.test.cjs
 PASS  tests/impactLogController.test.cjs
  impactLogController
    createImpactLog
      √ should create an impact log successfully with provided weight (5 ms)
      √ should use default category weight when weight is missing (1 ms)
      √ should use fallback weight 0.5 when category default does not exist (1 ms)
      √ should create Settings if none exist (1 ms)
      √ should return 400 if actionType or category is missing (1 ms)
      √ should return 400 for invalid actionType
      √ should return 500 on server error
    getImpactLogs
```

```
Backend > tests > settingsController.test.cjs > ...
12   describe("settingsController", () => {
75     describe("updateSettings", () => {
230      it("should return 500 if recalculation fails", async () => {
242
243        await updateSettings(req, res);
244
245        expect(res.status).toHaveBeenCalledWith(500);
246        expect(res.json).toHaveBeenCalledWith({
247          message: "Log fetch failed",
248        });
249      });
250    });
251  });
```

```
PS C:\Users\CHAMA COMPUTERS\OneDrive\Desktop\Eco-Project\project-EcoCycle\Backend> npx jest tests/holidayController.test.cjs
>> npx jest tests/settingsController.test.cjs

 PASS  tests/settingsController.test.cjs
  settingsController
    getSettings
      √ should return existing settings (4 ms)
      √ should create settings if none exist (2 ms)
      √ should return 500 on error (1 ms)
    updateSettings
      √ should update settings without recalculating logs if factor does not change (2 ms)
      √ should recalculate all impact logs if co2 factor changes (26 ms)
      √ should create settings if none exist before updating (3 ms)
      √ should update only monthly target when only monthlyTargetKg is provided (1 ms)
      √ should return 500 if settings save fails (1 ms)
      √ should return 500 if recalculation fails (2 ms)

Test Suites: 1 passed, 1 total
Tests:       9 passed, 9 total
Snapshots:   0 total
```